

Design of Local Area DGNSS Architecture to Support Unmanned Aerial Vehicle Networks: Concept of Operations and Safety Requirements Validation

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ABSTRACT

A local-area differential global navigation satellite system (LAD-GNSS) provides high integrity and accuracy for unmanned aerial vehicles (UAVs). In this study, three major issues were investigated to fully establish the LAD-GNSS and analyze the performance of the system. First, UAV operation concept and requirements are defined. This includes segregation of UAV operation coverage airspaces, and derivation of navigation requirements, alert limits (ALs), for each operational coverage. Second, LAD-GNSS architecture was designed by simplifying hardware and monitoring algorithms both on ground facility and onboard module using the well-established GBAS as a starting point. Lastly, theoretical performance evaluations were performed by comparing position uncertainty bounds, which are represented by protection levels (PLs), to the corresponding navigation requirements (ALs) for each coverages. PLs and ALs are derived and compared under both nominal and malfunction cases correspondingly. In addition, a method is described for deriving PLs for excessive acceleration and code-carrier divergence fault scenarios, which are bounded by using a maximum-allowable error in range. Using these PLs, the integrity/continuity allocations are ideally and dynamically assigned to each single-fault hypothesis to obtain optimized PLs that are identical for all fault scenarios. The results show that the VPLs are reduced by approximately 16% when implementing the optimal allocation method.

1. INTRODUCTION

One of the most essential technologies required for the operation of unmanned aerial vehicles (UAVs) is reliable and accurate computation of the navigation solution. The concept of a local-area UAV network that utilizes local-area differential global navigation satellite system (LAD-GNSS) solutions was proposed to achieve high navigation accuracy and reliability, which are required for autonomous UAV applications [1]. LAD-GNSS meets the required level of integrity requirements (comparable to those of GBAS Category I–III operations) by monitoring navigation faults at both the ground station and UAV and by broadcasting integrity information to the UAV. By utilizing the integrity information, which defines

navigation system error models, the UAVs can compute their conservative position error bounds (i.e., protection levels (PLs)) and consequently safe separation distances in real time. The LAD-GNSS architecture for the local-area UAV network proposed in previous work includes several methods of simplifying the current GBAS integrity monitoring algorithms and hardware configurations to lower the system cost and complexity while maintaining an acceptable safety level [1]. The performance and integrity of LAD-GNSS can also be improved by designing additional onboard modules. The LAD-GNSS integrity monitor testbed was installed on the KAIST campus. However, the above-mentioned modifications, including ground hardware/software changes and airborne components, have not yet been fully analyzed and implemented from a system-design perspective.

In this study, several key components of the LAD-GNSS architecture were developed, including system operation coverage and requirements, modifications of integrity monitors in both ground and airborne subsystems, and theoretical performance evaluations. The operation requirements, vertical alert limits (VALs), are derived based on the operating environment and uncertainty of navigation. The vertical protection level (VPL) under nominal and malfunction case, are compared to VAL to determine if the system meets the integrity requirements for a given operation.

The LAD-GNSS integrity monitors were developed using the well-established GBAS as a starting point. In this study, we implemented major modifications on the ground subsystems by designing additional onboard modules. First, instead of maintaining the ionospheric geometry screening at ground subsystem, which is computationally expensive due to computation of ionospheric error for the all-in-view geometry, the simplified airborne geometry screening are utilized. In this onboard module, an error induce by ionospheric anomaly is computed for a single geometry which UAV uses for its positioning.

In the current GBAS architecture, separate PL values for three scenarios (i.e., fault-free (H0), single reference-receiver fault (H1), and ephemeris-fault cases) are herein defined based on the predetermined integrity and continuity allocations. These allocations can ideally be dynamically assigned to make the PLs identical for all fault scenarios. The use of UAV information provides a

significant advantage over the conventional GBAS and enables allocations in real time. Since onboard module can know information about the satellite geometry utilized by the UAV for its positioning, it is possible to compute the PLs for all fault scenarios and optimally allocate the integrity and continuity budgets to make the PLs identical for all fault scenarios.

The remainder of this paper is organized as follows. Section 2 illustrates the LAD-GNSS architecture, including the operation coverage and integrity monitoring modifications. Section 3 explains the operation requirement of LAD-GNSS, and Section 4 details the method of theoretical performance evaluation. Section 5 concludes the paper.

2. LAD-GNSS ARCHITECTURE FOR UAV

The proposed LAD-GNSS architecture for UAV emphasizes low-cost and low complexity for commercial UAV applications with compact installation ground/onboard units while supporting high navigation accuracy and integrity within the coverage airspace.

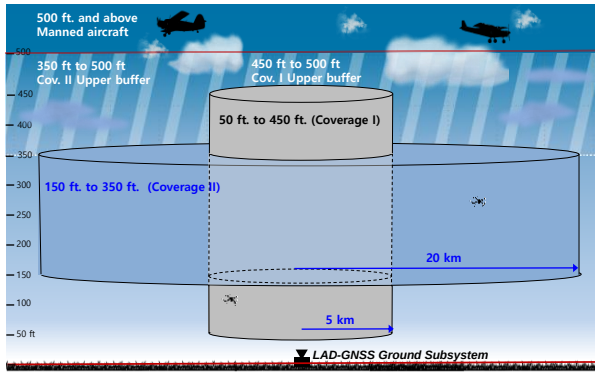


Figure 1. The concept of LAD-GNSS architecture for UAV (no ground obstacle is assumed)

Figure 1 illustrates the operational concept of LAD-GNSS architecture for UAV when there is no ground obstacles within operational regions. The ground facility shown at the center is the source of LAD-GNSS corrections and integrity information as well as real-time guidance for each UAV operating within the coverage airspace. To enable safe operations of UAV in the civil airspace, coverage airspaces were defined around the ground facility. The radius of coverage airspaces of the proposed system were defined based on expected UAV missions for each regions. ‘Coverage I’ supports low-speed UAV operations within the radius of 5 km from the ground facility. Coverage I is designed to potentially support UAV landing phase down to low limit of coverage I with high accuracy/integrity navigation performance. ‘Coverage II’ is designed to support high-speed operations, by having lower navigation accuracy/integrity compared to coverage I, within the radius from 5 to 20 km from the ground facility. The operation range of altitude for each coverage airspace are determined based on considerations of correction accuracy over the distance from the operating UAVs to the ground facility. Supported altitude for coverage I is defined

from 50 to 450 ft. This indicates that the upper and lower margins for vertical position uncertainty from both upper (500 ft) and lower (0 ft) hard bounds are considered as 50 ft. Supported range of altitude for coverage II is defined from 150 ft to 350 ft by assuming margins of 150 ft on both upper and lower sides. It should be noted that the upper and lower margins for coverage II is set larger than those for coverage I to take into account the degraded correction accuracy when the UAV operates at a longer distance.

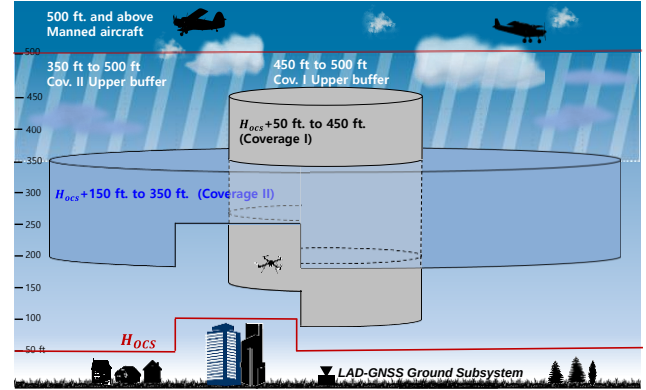


Figure 2 The concept of LAD-GNSS architecture for UAV (when H_{OCS} is assumed)

The operational concept introduced in Figure 1 can be applied in an ideal condition where there is no ground obstacles around operational regions. However, the assumption is not realistic for various commercial UAV operations. Thus, the concept of H_{OCS} is introduced. The H_{OCS} indicates the maximum height of an obstacle that can be placed along the aircraft approach path. H_{OCS} for GBAS have already been defined and are enforced at runways that have precision approaches, non-precision approaches and even visual operations [2]. Since the environments that UAVs fly above is more variable compared to those for manned aircraft, H_{OCS} for UAVs should be adaptively defined based on the environments where the UAV is actually flying above. This H_{OCS} can be pre-defined prior to operation based on environmental conditions. It is noticeable that the operational concept introduced in Figure 1 is a special case of operational concept when H_{OCS} is 0. Figure 2 shows the operation concept with pre-modelled H_{OCS} . H_{OCS} is modeled to bound the heights of buildings and trees in ground that should not be violated for safety. By assuming the H_{OCS} , the operational range of altitudes become variable and reduced. The defined coverage airspaces are summarized in Table 1.

Table 1. Coverage airspace of LAD-GNSS

Criteria	Radius (from the ground facility)	Altitude
Coverage I	~ 5km	$50\text{ft} + H_{OCS}\text{ ft} \leq h_1 < 450\text{ ft}$
Coverage II	5km~20km	$150 + H_{OCS} \leq h_2 < 350\text{ ft}$

Figure 3 describes the simplified configuration and components of LAD-GNSS that includes the ground facility (Fig. 3(a)) and the UAV onboard module (Fig. 3(b)). Three ground reference receivers are connected to

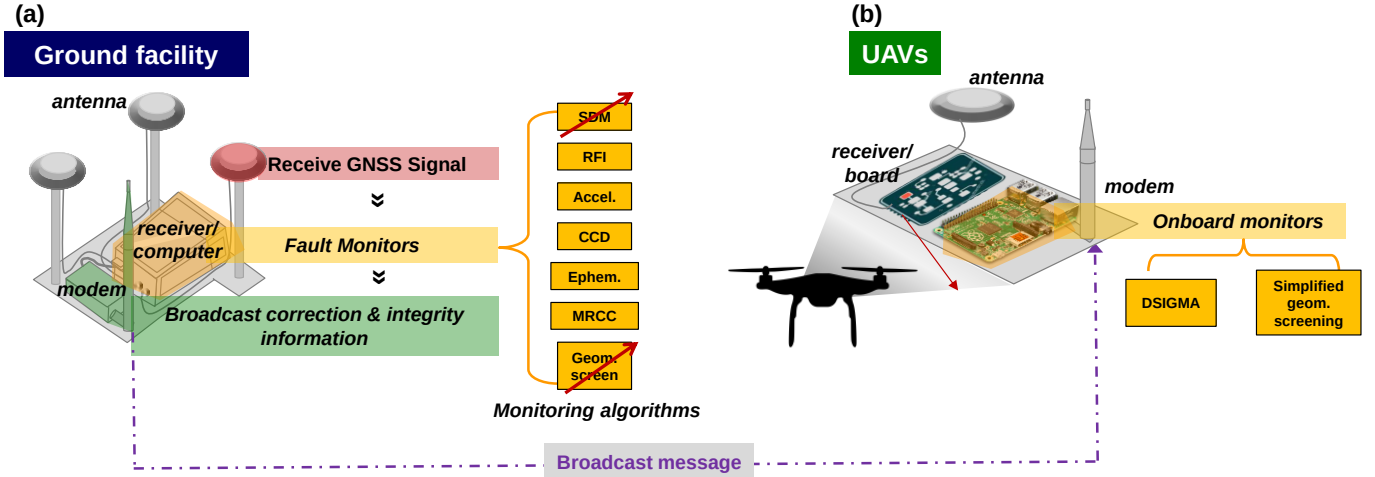


Figure 3. The configuration and components of LAD-GNSS architecture

antennas are sited close together. Low-quality antennas and relatively short antenna separation increase the impact of multipath errors. To mitigate performance degradation due to the hardware restriction, and to take advantages from the new system configuration, this study develops the several key components of the LAD-GNSS architecture including optimal integrity allocations and the modifications of integrity monitors. The optimal integrity allocations allocate total system integrity risk budget dynamically to make the final position uncertainty bounds identical for all fault modes. This would reduce final user position uncertainty bound under nominal case compared to the original bound computed from the pre-allocated integrity budgets. The algorithm for optimal integrity allocations is introduced in Section 4. In addition to the optimal allocation of integrity budget, modification of monitors can potentially improve the overall performance. To mitigate the integrity risk due to anomalous ionosphere gradients, DSIGMA monitor (which is adopted for GBAS CAT-II/III) is additionally employed with the use of a code-carrier divergence (CCD) monitor to meet the system requirement. In addition, geometry screening is moved from ground facility to onboard to reduce the conservatism and computational burdens based on known user satellite geometry on onboard. The simplified geometry screening computes position uncertainty under ionospheric anomaly condition in real-time for only a single satellite geometry that user is tracking instead of computing impacts for all the possible user geometries. The concept of modified geometry screening and position uncertainty simulations will be discussed in Section 5. Lastly, Signal Deformation Monitor (SDM) also can be eliminated by selecting ground and airborne GNSS receivers that are either the same or very similar. This would cancel out the resulting error when users apply their differential corrections [1].

3. OPERATION REQUIREMENT FOR LAD-GNSS FOR UAV

low-cost antennas. By considering siting constraints for most commercial applications will not have room to spread ground antennas over hundreds of meters,

PLs are compared to alert limit in each position axis of interest to determine if the system meets the integrity and continuity requirements for a given operation. In this paper, the vertical alert limits (VALs) of Coverage I and II are analyzed based on the total system error (TSE), specifically its composition of the sum of the navigation sensor error (NSE) and flight technical error (FTE). It is sufficient to consider only the vertical requirements since they are more difficult to fulfill than the lateral ones.

The key requirement for vertical TSE performance of LAD-GNSS is ensuring actual UAV location should not violate the H_{ocs} or manned aircraft zone. For example, the vertical TSE of a UAV operating in Coverage I must not exceed 50 ft because the low limit of Coverage I is defined as 50 ft + H_{ocs} . Corresponding to these requirements, the analysis is divided into a nominal NSE case, followed by a malfunction NSE case. The nominal case represents that no faults occur (i.e. fault-free condition) or faults detected by monitors in the navigation system (faulted condition), and the malfunction case represents that an unmonitored fault occurred that caused a large position error.

3.1 Vertical Alert Limits under Nominal Case

In analysis of the nominal NSE case, allowable probability of exceeding the limit for the fault-free condition is 10^{-6} ; Allowable probability of exceeding the limit for the faulted condition is 10^{-5} . The paper handles only the fault-free condition for VAL of nominal NSE case ($VAL_{nominal}$) because it has a tighter requirement than that of the faulted condition. To facilitate the analysis, simplified assumptions are used in this analytical derivation. It is assumed that the relationship between vertical NSE and $VAL_{nominal}$ at the worst geometry is expressed as

$$\sigma_{NSE,vert}(m) = \frac{VAL_{nominal}(m)}{k_{FFMD}}; \quad k_{FFMD} = 5.43 \quad (1)$$

where k_{FFMD} is a multiplier defined to ensure a specified probability of fault-free missed detection for the case of three ground reference receivers. The vertical TSE is normally distributed with a zero mean and standard deviation ($\sigma_{TSE,vert}$) given by

$$\sigma_{TSE,vert} = \sqrt{\sigma_{NSE,vert}^2 + \sigma_{FTE,vert}^2} \quad (2)$$

It is assumed that, if TSE is zero, the UAV height will be low limits (50 ft from H_{OCS} for the Coverage I or 150ft from H_{OCS} for the Coverage II). Therefore, the probability that the height of UAV will be lower than H_{OCS} is given by

$$P_{risk,Coverage I} = \int_{50}^{\infty} \frac{1}{\sqrt{2\pi}\sigma_{TSE,v}} e^{-\frac{x^2}{2\sigma_{TSE,v}^2}} dx \quad (3)$$

$$P_{risk,Coverage II} = \int_{150}^{\infty} \frac{1}{\sqrt{2\pi}\sigma_{TSE,v}} e^{-\frac{x^2}{2\sigma_{TSE,v}^2}} dx$$

Figure 4 show plots of the fault-free risk versus $VAL_{nominal}$ for values of $\sigma_{FTE} = 3\text{ m}$, $\sigma_{FTE} = 2.7\text{ m}$, $\sigma_{FTE} = 2\text{ m}$, and $\sigma_{FTE} = 1.5$. Previous works have shown that σ_{FTE} of the multi-copter UAV is a sub-meter level [5]. Based on the conservative value of σ_{FTE} ($= 1.5\text{ m}$), the $VAL_{nominal}$ of 15.3 m for Coverage I and the $VAL_{nominal}$ of 51.4 m for Coverage II are determined.

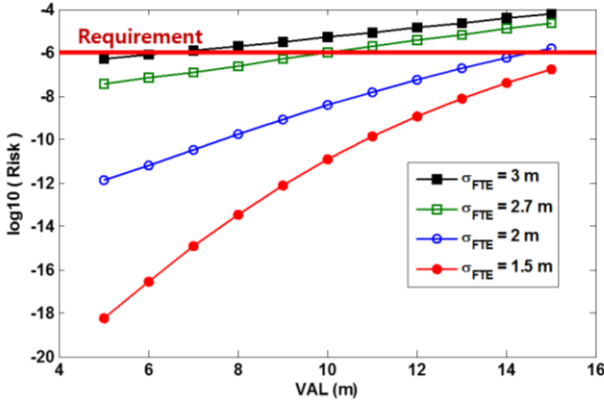


Figure 4. Fault-free risk for Coverage I

3.2 Vertical Alert Limits under Malfunction Case

An undetected propagation anomalies such as those induced by abnormal ionospheric or tropospheric conditions are considered as malfunctions. These errors resulted from malfunction is assumed as a bias. Therefore, the sum of the bias due to the malfunction and nominal error term must not exceed position uncertainty budgets (50 ft for Coverage I or 150 ft for Coverage II). In the malfunction case, the nominal errors are fixed at their 95th percentiles as shown in Eq (4).

$$VAL_{malfunction,Coverage I} + TSE_{95\%,vert}(D) \leq 50\text{ ft} \quad (4)$$

$$VAL_{malfunction,Coverage II} + TSE_{95\%,vert}(D) \leq 150\text{ ft}$$

where D is distance between the ground facility and the UAV. $TSE_{95\%,vert}$ is given by

$$TSE_{95\%,vert} = 1.96 \sqrt{\sigma_{NSE,vert}^2 + \sigma_{FTE,vert}^2} \quad (5)$$

In this study, the $TSE_{95\%,vert}$ computed by using the σ_{FTE} of 1.5 m and σ_{NSE} of 1.7 m. Using the Eq. (4) and (5), the $VAL_{malfunction}$ of 10.8 m for Coverage I and $VAL_{malfunction}$ of 41.5 m for Coverage II are determined. As the VALs derived from malfunction case are smaller than the $VAL_{nominal}$ in both coverages, we selected the $VAL_{malfunction}$ as the final VAL for each region.

4. THEORETICAL PERFORMANCE EVALUATION

In the LAD-GNSS for UAV, two VPLs are computed in real time. The first one is VPL for nominal case that represents fault-free condition or faulted condition. By taking benefits from the known satellite geometry, $VPL_{nominal}$ with optimal allocation of integrity and continuity is computed. The other one is VPL for malfunction that represent the position error due to undetected ionospheric anomaly. Since the ionospheric anomaly is a representative malfunction case in navigation system error, only $VPL_{malfunction}$ in ionospheric anomaly case is computed. The resulting VPL_{final} is determined by taking maximum among those two VPLs. When the VPL_{final} is less than or equal determined VAL in Section 3, UAVs operation in each coverage is available.

4.1 Vertical Protection Level under Nominal Case

In the current GBAS architecture, separate PL values for fault-free, single reference-receiver fault, and ephemeris-fault cases are defined based on the predetermined integrity and continuity allocations. For the rest of those fault modes, ground facility ensure that position uncertainty due to other fault mode not exceed VPL_{H0} . This concept is commonly known as a maximum-allowable error in range (MERR). This causes the conservative VPL because they may hold excess integrity allocation. As mentioned in Section 1, using the known satellite geometry of UAV, the onboard module can compute the PLs for all fault scenarios and optimally allocate the integrity and continuity budgets to make the PLs identical for all fault scenarios.

Figure 5 show the integrity fault tree of the system to which the optimal allocation was applied. The total integrity risk requirement of 2×10^{-6} per 150 is allocated. As UAV accidents can be expected to have less casualties than manned aircraft accidents, the relaxed value of the GBAS CAT I is allocated. In addition, sub-allocations are adjusted by the inclusion of the optimal allocation. In the re-designed integrity fault tree, the integrity risk of single deformation, low signal level, and all other failures are pre-

allocated. This is because that the signal deformation error is canceled out when the ground subsystem and UAV use the receivers that are either the same or very similar. Thus, the small value (1×10^{-8} per 150 s) is allocated to the signal deformation integrity risk. In the case of the low signal level integrity risk, the 1×10^{-8} per 150 s is also pre-allocated because the bound of the low signal level is determined based on the impact on the performance of other monitors, not on MERR. Finally, as all other failures are bounded by using the $VPL_{malfunction}$, the integrity risk, which is the same as that of GBAS CAT I, is allocated.

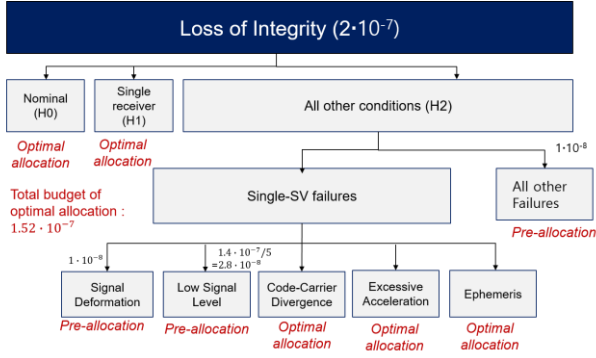


Figure 5. Integrity risk allocation of LAD-GNSS for UAV

As the allocations are optimized with the PLs for each fault mode, the PLs for fault modes that were bounded by using the MERR method should be newly derived. The VPL under the fault hypothesis for satellite i consists of two components, as shown in Eq. (6). First, the position error bound (μ_{fault}) is computed based on the bias induced by the satellite fault. Second, the random error term is computed using a sigma-multiplier (k_{HMI}), which considers that monitors miss-detect such faults. As the μ_{fault} is depend on the fault hypothesis, it is important to find relationship between the monitor domain errors and user range domain errors to derivate the VPLs for fault mode. In this paper, the VPLs for excessive acceleration (EA) and code-carrier divergence (CCD) are developed.

$$VPL_{fault}(i) = \mu_{fault,i} + k_{HMI,EA} \sum_{i=1}^n S_{vert,i}^2 \sigma_i^2 \quad (6)$$

where k_{HMI} is a multiplier defined to ensure a specified probability of the missed detection, and σ_i is the standard deviation of the differential ranging error for satellite i . Additionally, S_{vert} is the row of the weighted-least-squares projection matrix corresponding to the vertical position state [3].

5. Conclusion

In this paper, we presented the LAD-GNSS architecture for UAVs that communicate with ground subsystems in real time. LAD-GNSS for UAV was fully analyzed and implemented. It consists of an operation

requirement design, integrity monitoring strategies, and theoretical performance evaluation. The operation coverages that were defined for two regions have respective operating requirements. For Coverage I, VPL was compared to VAL of 10.8 m which is determined by analyzing uncertainty of navigation in nominal and malfunction condition. The UAV operating Coverage II was expected to meet the VAL of 41.5 m. In this paper, we show that the final VPL in Coverage I is determined from the $VPL_{nominal}$, while, in Coverage II, the final VPL is determined from the $VPL_{nominal}$.

In addition, we derived the PLs for EA and CCD fault cases to implement the optimal allocation of integrity/continuity budgets. We simulated the VPL with an optimal allocation. In the simulation results for a 24-satellite GPS constellation, VPLs of Coverage I and II were reduced by approximately 16% when implementing the optimal allocation method.

The NSE and FTE errors highly depended on the hardware, environment, and operational conditions and thus needed to be experimentally evaluated under various conditions. To this end, future work will include both the simulation-based analysis and field tests of several types of UAVs in different flight environments and modes.